The Japan Society of Mechanical Engineers

Proceedings of ICONE-23
23rd International Conference on Nuclear Engineering
May 17-21, 2015, Chiba, Japan

ICONE23-1131

THE EUR COMPLIANCE ANALYSIS OF THE EU-APWR STANDARD DESIGN: ASSESSMENT PROCESS, METHODOLOGY AND MAIN RESULTS

Luca Facciolo
Vattenfall AB
Solna, Sweden

Pekka Nuutinen
Fortum
Espoo, Finland

Daniel Welander
Vattenfall AB
Solna, Sweden

Keywords: European Utility Requirements, EUR Document, EU-APWR, Design Assessment.

ABSTRACT

The European Utility Requirements organization started the compliance assessment process of the Mitsubishi Heavy Industries EU-APWR Standard Design in 2012.

The EU-APWR is an Advanced PWR, 1700 MWe class, 4-loops, 14ft active core fuel length that MHI has developed for the European market. The EU-APWR is an evolution of the Advance PWR currently under the licensing process in Japan for the Tsuruga Power Station. MHI has modified the design applying improvements in safety and economy in order to be adapted for European markets and to comply with the EUR requirements.

The EU-APWR Standard Design documentation has been assessed against the EUR Volume 2 – Generic Nuclear Island requirements Revision D, issued in October 2012. The assessment is divided into 20 chapters for a total of over four thousand individual requirements.

Each chapter was assigned to Assessment Performers who executed the detailed analysis of the requirements. The assessment of each requirement and the Synthesis Report have been submitted to, and scrutinized by, the Coordination Group, formed by representatives of the European Utilities together with the Vendor, and reviewed by the Administration Group and by the Steering Committee.

The Synthesis Reports have been collected in the Volume 3 EU-APWR Standard Design Subset and presented to the Steering Committee, which approved the final draft in October 2014.

The overall results of the assessment indicated good compliance of the EU-APWR Standard Design: 77% of the requirements resulted in compliance with EUR. This percentage increases to 85% when taking into account the requirements for which the design has been evaluated in compliance with the objectives. The requirements where the design has been judged not in compliance with EUR are less than 2%.

The divergences between the EU-APWR Standard Design and the EUR concern different areas like, for instance, layout, operational capability and performance, outage operations, personal protection and radiation monitoring.

Some of the incongruences result from differences in approach to the design process or from differences in the rules and standards in use in Japan and in Europe.

Some analyses, like the internal hazards effects, have been performed only partially because, in Japan, such analyses are considered site-specific and are carried out at the detailed design level.

The analysis of the consequences of a hydrogen explosion, and the environmental qualification methodology of equipment have not been fully developed yet.

While the reactor core has been designed for an operability cycle of 24 months and can be loaded with 50% MOX fuel, no other area of the plant has been designed taking into consideration MOX fuel.

1. INTRODUCTION

The scope of the EUR organization is to build a reference technical document that could be used by most European Utilities for the development of possible new Nuclear Power Plants and for the bidding of new Nuclear GEN 3. The EUR wants to promote safe and economical new nuclear in Europe pushing the Regulators and the Vendors to increasingly standardized and safe design taking into account all the lessons learnt from the nuclear industry.

In general the project helps in improving the understanding of different cultures, in developing engineering tools and skills, in increasing the Utilities´ knowledge of the Vendors'

Copyright © 2015 by JSME

NII-Electronic Library Service

The Japan Society of Mechanical Engineers

designs and, on the other hand, the Vendors' understanding of the EUR documentation.

Mitsubishi Heavy Industries contacted the EUR organization asking for the elaboration of the EUR Volume 3 subset H EU-APWR Standard Design. The EUR organization and the Utilities expressed their interest in the project so that the Steering Committee (SC) approved the Vendor´s (MHI) request in 2011. A Coordination Group (CG) was set up in 2012 in order to administrate the work.

The EU-APWR is an Advanced PWR, 1700 MWe class, 4-loops, 14ft active core fuel length that MHI has developed for the European market. The EU-APWR is an evolution of the Advance PWR currently under the licensing process in Japan for the Tsuruga Power Station. The Japanese version was first adapted for the American market, then adapted for the European market. MHI modified the design applying improvements in safety and economy in order to comply with the EUR.

The EU-APWR is the first design to be assessed against EUR Volume 2 – Generic Nuclear Island requirements Revision D, issued in October 2012 [1]. This volume contains generic requirements that are not related to any specific design. Volume 2 is divided into 20 chapters which cover the whole scope of the activities necessary in order to design, license, build, test and operate the Nuclear Island efficiently and safely for a total of over four thousand individual requirements, see chapter division in Figure 1.

VOLUME 2 : GENERIC NUCLEAR ISLAND REQUIREMENTS	
Chapter 2.1	Safety requirements
Chapter 2.2	Performance requirements
Chapter 2.3	Grid requirements
Chapter 2.4	Design basis
Chapter 2.5	Codes and standards
Chapter 2.6	Material-related requirements
Chapter 2.7	Functional requirements: components
Chapter 2.8	Functional requirements: systems & processes
Chapter 2.9	Containment system
Chapter 2.10	Instrumentation-and-control and Human-machine interface
Chapter 2.11	Layout
Chapter 2.12	Design process and documentation
Chapter 2.13	Constructability & commissioning
Chapter 2.14	Operation, maintenance and procedures
Chapter 2.15	Quality assurance
Chapter 2.16	Decommissioning
Chapter 2.17	PSA methodology
Chapter 2.18	Performance assessment methodology
Chapter 2.19	Cost assessment information requirements
Chapter 2.20	Environmental impact

Figure 1 EUR Volume 2 Chapter division

2. PREPARATORY PHASE

The initial process includes the preparation and the acceptance of the EU-APWR Standard Design documentation, which is divided into the EU-APWR Plant Description Documents and the EU-APWR Plant Review Package. All the documents are based on a frozen design that means it is not possible to modify the description of the design during the assessment work.

The EU-APWR Plant Description Documents include:

- Technical Plant Description, a summary of the frozen EU-APWR design available to all EUR members,
- Design Description Documents, an explanation of the basic design of EU-APWR, only available to CG members.

Before the assessment of each EUR Volume 2 chapter, MHI provided the EU-APWR Plant Review Package which consists of:

- Self-assessment sheets, indicating what reference document to use and what section to study, suggesting a rationale and an assessment result for each requirement,
- Supplementary documents, all the support documentation used in the self-assessment and not already delivered, available to all CG members.

The self-assessment sheets are not formal design documents: they are only used as an input to facilitate the work of the Assessment Performer.

The Supplementary documents could be the results of the response to an official technical question or they could include detailed design information on the US-APWR design, which is the reference plant for the EU-APWR.

3. ORGANIZATION

The project organization is presented in Figure 2. The EUR utilities that volunteered to take part in the assessment work are divided into Sponsors, who are responsible for the production of the detailed Analysis of Compliance Reports, and Supporters, who review the reports.

The EUR utilities together with the Vendor and the Project Management Team (PMT) constitute the Coordination Group (CG) which administers the project and is responsible for its development and results.

Each EUR utility appoints a representative, who acts as contact person in the CG.

The assessment work is done by a team of Assessment Performers coordinated by the Chapter Leader, appointed by the Sponsor, who is responsible for the technical work and the process of generating the Analysis of Compliance Reports to be presented during the CG and AG (Administration Group) meeting (see section 4.2).

The Chapter Reviewer, appointed by the Supporter, is in charge of reviewing the work done by the Assessment Performers.

The Project Management Team is responsible for the organizational and administrative matters and represents the interface between the CG and the AG/SC.

MHI, the EU-APWR Vendor, is responsible for the self-assessment, the delivery of all the necessary and required documentation and the technical support to the assessment.

The Administration Group plays the role of project sponsor and project steering group. It advises the project on strategically important issues giving guidance and experience from earlier EUR projects and is responsible for the review of the Compliance Assessment and its consistency with previous assessments of other designs.

Copyright © 2015 by JSME

NII-Electronic Library Service

The Steering Committee is the owner of the project and makes the final statement regarding the administration and the assessment, being in charge of managing all the topics that cannot be solved at the AG level.

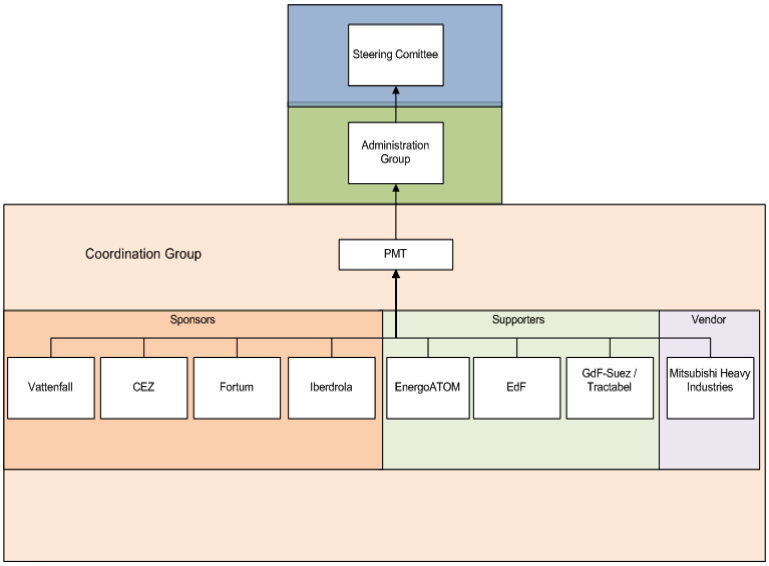


Figure 2 EUR-organization

4. COMPLIANCE ASSESSMENT PROCESS AND METHODOLOGY

The execution phase of the project consists in the technical assessment of the EU-APWR standard design.

The Assessment Performers are in charge of carrying out the Compliance Assessment based on the EU-APWR Design Documentation and of presenting the Analysis of Compliance Report for each chapter.

The assessment process is divided into different steps with an iterative methodology (see Figure 3).

The main steps in the methodology are:

- Start-up of a Chapter Assessment
- Performing the technical assessment
- Compiling the Analysis of Compliance Report
- CG review meeting
- AG and SC review meeting
- Generation of EUR Volume 3 subset H EU-APWR

4.1 Performing the Assessment

The EUR Secretariat is responsible for providing the Assessment Sheets, which are the template for the Analysis of Compliance Report and are divided into four columns.

The first column is the only one already filled in and it contains the EUR Requirements. The Requirements are marked by the word “shall” or “should” in blue letters and cannot be modified, unless there are mistakes. The Assessment Performer shall check the compliance with all the explicit requirements presented in the first column.

In the first column are also presented the possible Comments, written in italics, used to clarify the requirement. The comment cannot contain any “shall” or “should”.

The second column is reserved for the References. A detailed reference to the official documentation (document number, section, page) needs to be provided in for each assessed requirement.

The third column is dedicated to the compliance assessment. It contains a comprehensive description of the strategy used in the EU-APWR to deal with the topics and the rationale of the assessment.

The fourth and last column is used for the results of assessment. A label is assigned to each requirement. The labels and their meaning are listed in Table 1.

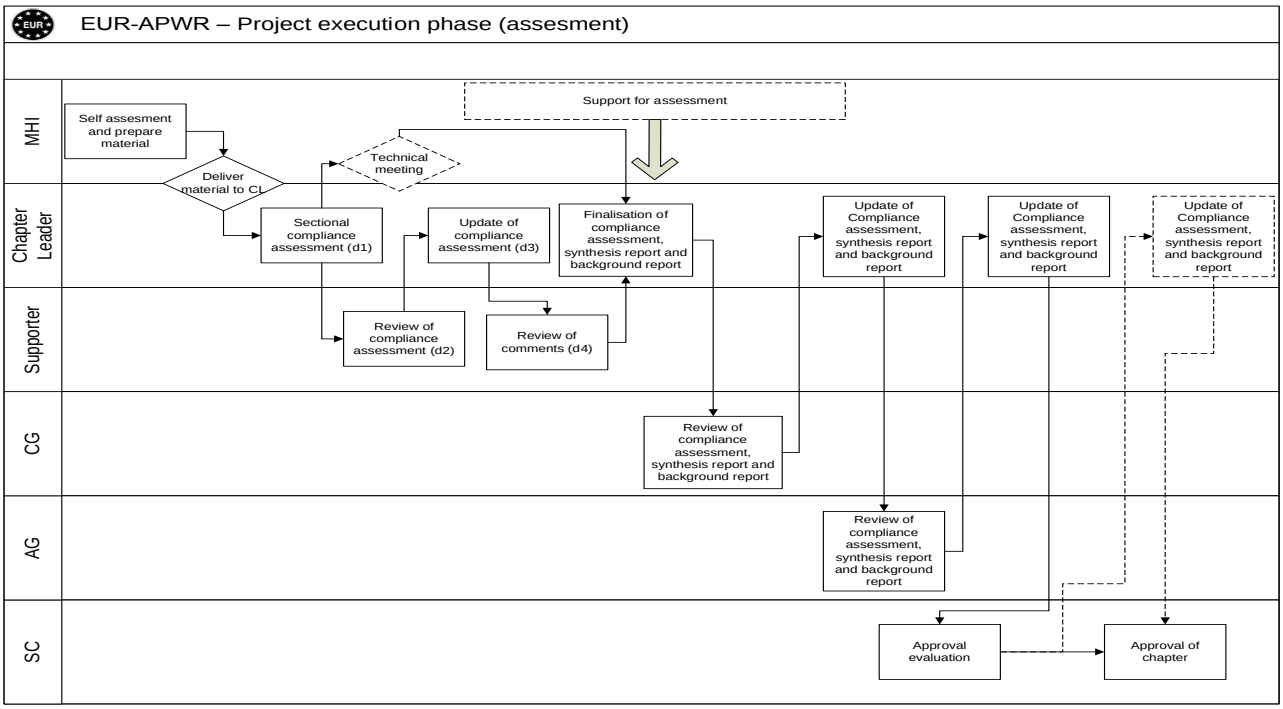


Figure 3 Project execution phase, assessment

Copyright © 2015 by JSME

NII-Electronic Library Service

The Japan Society of Mechanical Engineers

Compliance Assessment Label	Meaning	Classification Label
compliance	the design meets the requirement but does not go significantly beyond	COM
compliance with objectives only	the design is supposed to achieve the objective of the requirement, but: A) a different approach from the EUR one is used to achieve the same objectives, or B) the approach is not yet sufficiently defined for a COM, but there is a fair expectation based on provided information and experiences that the Vendor will fulfil the requirement in the later phase of the design.	CWO
non compliance	the design does not meet the requirement	NOC
significantly beyond requirement	the design significantly exceeds what is requested by EUR	BRE
not applicable	the requirement is not applicable to the design	NAP
not assessable today	the assessment cannot be made for some reason (e.g. basic design vs. detailed design, or standard vs. site specific issues)	NAS
see another section	the requirement is assessed in another section; reference must be given	SEE
background note	a proposal for change to the EUR text is given	HOLD
information needed	additional information is expected from the vendor or EUR before completion of the assessment	INF

Table 1 assessment labels

The CWO label is to be used if the objective of the requirement is achieved following a different approach from the one indicated by EUR or if engineering judgment indicates the expectation that the Vendor will be able to satisfy the requirement at a later stage of the design.

The BRE label indicates that the design not only meets, but significantly exceeds what is required. This could be interpreted both as a positive label, indicating a wide margin in the design or as a negative one, indicating over-designing.

The label NAP is used for technological solutions that do not apply to the design, for instance BWR-specific requirements do not apply to PWR designs.

The label NAS is used in case a site, an owner or a specific project is needed and when the design or the verification will be developed at a later stage in the design process and it is considered acceptable.

The SEE label is used to make reference to another requirement where the subject has already been discussed. Since the same topic could be addressed in different requirements in different levels of detail and in different chapters, it is necessary to perform a cross-reference check in order to avoid inconsistencies of conclusion or mutual references ending with the requirement not being assessed anywhere.

During the Compliance Assessment a requirement can be temporarily marked with the INF label. This indicates the necessity to have additional information and technical support from the Vendor in order to correctly evaluate the topic. The Chapter Leader is in charge of starting a Question and Answer procedure where the technical questions are prepared in a Q&A-template and sent to the Vendor. The Q&A procedure could be an iterative process until the Chapter Leader formally accepts the answer and the document becomes an official document that can be included as reference in the second column of the Assessment Sheet.

It is possible for the Vendor to update the Design Description Documents if the Q&A process shows the necessity of more detailed information or an obvious error in the documentation provided. In this case the Vendor issues a new revision of the document with clear identification of the changes.

At the end of the Compliance assessment the Chapter Leader is in charge of writing up the Synthesis Report, which is a self-standing document reporting the conclusions and the summary of the assessment work. The Synthesis Report content for each chapter is:

1. Summary of Compliance Assessment
   1.1. Qualitative Statement of the Assessment
   1.2. Indicative Statistics
2. Main CWO
3. Main NAS
4. Non-compliance (NOC)
5. Beyond requirement issues (BRE)
6. List of references

The Synthesis Report is the basis for the Subset H, chapter 2 “Highlights of Compliance Analysis” [2].

4.2 Review of the Compliance Assessment

The assessment work done by the Chapter Leader is first reviewed by the Chapter Reviewer who has the responsibility of checking the rationale and the references for every requirement assessed. This task is performed during the chapter assessment period in order to have an efficient time schedule and it implies a continuous interaction between the Chapter leader and the Chapter Reviewer. If there is divergence of opinion regarding some requirements these are discussed during the CG meeting.

The Analysis of Compliance Report is delivered to the CG at least two weeks before the meetings so that the CG members have the time to review it. Due to the lack of time during the meetings, the discussions are restricted to the following cases:

Copyright © 2015 by JSME

NII-Electronic Library Service

The Japan Society of Mechanical Engineers

- Consensus was not reached between Assessor and Reviewer,
- There is a need for advice from CG,
- Label is other than COM, HOLD,
- CG-members want to discuss the requirement assessment for whatever reason.

The CG is in charge of pointing out which requirements need advice from the AG.

The Chapter Leader updates the Analysis of Compliance Report and the PMT delivers the document to the AG at least two weeks before the meeting. The requirements to be discussed during the AG meeting are highlighted, and concern:

- Requirements where consensus was not reached in CG,
- Requirements related to unique features of the design or where the compliance assessment is considered controversial,
- Requirements with the labels NOC, BRE, CWO and NAS,
- Any other requirements that individual AG-members wants to discuss,
- The Synthesis Report.

If no consensus can be reached on an issue during the AG meeting, it is marked as open item and it is passed to the next AG meeting with a suggestion on how to close it. The AG should try to find consensus on all controversial issues. Anyhow, if it is not possible, the SC will take the final decision.

It can happen that an INF label cannot be resolved before assessing another chapter. The PMT should take note of such INF labels and ensure that the corresponding requirements are assessed when the information becomes available.

After the AG meeting, the Chapter Leader updates the assessment and the PMT distributes the documents to the SC for review.

During the SC meetings only high-level issues and the Synthesis Report will be discussed, mainly concentrating on the requirements labeled as NOC and CWO and the requirements for which the AG asked for advice.

The document is then updated with the SC comments and officially approved during the following SC meeting.

The final versions of the EU-APWR Compliance Assessment and Synthesis Reports were approved by the Steering Committee in October 2014.

5. MAIN RESULTS

The overall results of the assessment process indicate good compliance of the EU-APWR Standard Design with the EUR requirements: 76.8% of the requirements resulted in compliance (COM) with EUR. Taking into account also the requirements in compliance with the objective (CWO), the percentage increases to 84.8% (excluding the NAP and SEE labels). The majority of CWO labels are due to items of the design not being sufficiently defined at that stage.

The EU-APWR Standard Design studied for the Compliance Assessment was at the basic design phase. Since the safety concept and the plant layout are not the same, it was not possible to fully use the documentation applicable to the reference design US-APWR during the assessment. As a consequence a large part of the requirements was marked with the NAS label, 13.3%.

The requirements not in compliance (NOC) amount to 1.3%. This confirms the good compliance. It has to be noted that no NOC label was considered so serious as to hinder a plant project in Europe.

As explained before, not all the requirements are directly applicable to the EU-APWR Standard design, partially due to the technology (like BWR-related requirements) and partially due to the technical features (like the Low Head Safety Injection Pumps, which in the EU-APWR are replaced by the advanced accumulator). The total percentage of NAP labels is 6%.

In the following sections some of the main items are presented according to their respective label.

5.1 Main CWO

A great number of requirements are labeled CWO because the approach used by MHI is not sufficiently defined even if engineering judgment on the provided information can demonstrate sufficient confidence in the ability of MHI in solving the issue respecting the requirement.

The EU-APWR Standard Design corium cooling strategy is based on ex-vessel cooling of fragmented solidified corium on cooling elements in the Reactor Cavity. Below the reactor pressure vessel is situated a large steel bowl device (MFEED) with a small hole in the bottom which collects and discharges the corium in a controlled way to the Reactor Cavity. Assuming a global vessel failure, the vessel wall will creep and the vessel bottom may be repositioned on the MFEED blocking the hole and preventing its operation. MHI did not present any analysis of the behavior of the melt in such situation, as no creeping was assumed for the analysis of the melt flow into the MFEED and through the hole. Further analysis may be able to show the independence of vessel failure mode.

The severe accident termination concept of the EU-APWR involves fully filling the reactor cavity with water before vessel melts through. Corium coolability is demonstrated in an analysis where both calculations and experiments are taken into account. If the debris is sufficiently cooled in the Reactor Cavity no concrete erosion will occur. The Probabilistic Safety Analysis (PSA) also shows that no erosion will occur when the Reactor Cavity is filled with water. However it is impossible to avoid the full erosion of the concrete reactor cavity floor in the case of Severe Accident characterized by a dry Reactor Cavity. MHI has calculated a very low probability for the Severe Accident scenarios with dry reactor cavity that will lead to the concrete erosion. The probabilities are considered as a residual risk.

The global hydrogen detonations are prevented by the use of large containment volume, sufficient mixing, and the installation of Passive Autocatalytic Recombiners (PAR). The calculations run with a lumped parameter code cannot totally exclude the occurrence of local hydrogen detonation

Copyright © 2015 by JSME

NII-Electronic Library Service

The Japan Society of Mechanical Engineers

when using conservatively selected criteria. No further detailed analysis of flame acceleration behavior has been performed yet. MHI has committed to perform more detailed analyses and to take the necessary precautions in case the analyses show the possibility of local hydrogen detonation.

The seismic design of the EU-APWR Standard Design applied the spectra based on RG 1.60, with some modifications, which results in an enveloping spectrum of the three spectra of EUR. The approach used by MHI is considered acceptable. Still, it is not possible to demonstrate it is consistently conservative without the details of the calculations because it is different in principle from the EUR requirement.

The EUR states that the Safety Injection System (SIS) injection pressure should remain below the setpoint of the Steam Generator secondary side safety valves in order to minimize containment bypass following a Steam Generator Tube Rupture (SGTR). In the EU-APWR Standard Design the set point of the Main Steam Safety Valve is 8.17 MPa (gauge) while the shut off pressure of the Safety Injection pumps is 13.5 MPa (gauge). However, in the case of SGTR, the abnormally high Steam Generator water level signal will open the Steam Generator water discharge valve (SGWDV) automatically releasing the Steam Generator water into the refueling water storage pit (RWSP) which is located inside the containment. The SGTR safety analysis report shows that it is not possible to completely avoid the containment bypass but the design is intended to minimize it.

The Design Manual that should contain the design plan and the design process wasn't provided by MHI. The design documentation shows only the outline of the future Configuration Management system stating that the main principles and the main procedures will be written according to the following nuclear specific standards:

- IAEA Safety Standards GS-R-3 "The Management System for Facilities and Activities" (2006),
- ANSI/NIRMA CM1.0 "Guideline for Configuration Management* of Nuclear Facilities" (2007).

5.2 Main NOC

Almost half of the internal hazards events listed by EUR to be considered for the pressure retaining parts have not been taken into account. MHI states that the remaining internal hazards will be analyzed at a later stage in a detail design specific project. EUR's position is that such events should be taken into account at the basic design stage since the analysis results could have a major impact on the design that could be difficult to implement at the later detailed design phase.

MHI's analysis results for deflagrations to detonation transition and flame acceleration show that the limits are violated during a short period of time. For this reason MHI assumes that no local hydrogen explosion will occur and no analyses of the consequences of such explosion have been performed (see previous section Main CWO). Therefore, uncertainties exist concerning the impact from local detonations on important internal structures, systems and components and the secondary impact on the integrity of the containment from missiles generated by a local detonation.

MHI performed probabilistic safety analysis (PSA) for groups of active, redundant and same type of components subject to Common Cause Failure (CCF), but the CCF is considered only within the same system and not between different systems as explicitly required by EUR.

The EU-APWR Standard Design requires an adjustment of the soluble boron concentration during Load Following. Even if the concentration change needed is marginal, EUR requires the achievement of Load Following without adjusting soluble boron concentration during the manoeuvre.

The refueling outage is estimated to last 19.8 days for the EU-APWR Standard Design. Although this duration is based on Japanese regulator procedures and MHI performed an analysis showing the possibility to shorten the outage duration, it cannot reach the EUR limit of 14 days.

Further the time from hot shut down conditions to generator synchronization is approximately 46 hours, based on Japanese experience, possibly 40 hours if some physic tests are shortened, which is nevertheless above the 24 hours specified by EUR.

The core of EU-APWR Standard Design can accommodate 50 % MOX fuel, but the adoption of MOX fuel assemblies is not taken into consideration in the design of other areas of the plant. Even if it could be anticipated that MHI is able to design the storage facility for MOX fuel, the standard EU-APWR Standard Design fuel storage facility, especially the new fuel storage facility, has not taken into consideration the use of MOX fuel.

MHI did not develop the environmental qualification sequence for the EU-APWR Standard Design. Since the component choice is done at the later stage of the detailed design, EUR considers that this issue should be resolved at the basic design phase in order to guarantee which qualification level will be required for the different safety system components concerning aging, irradiation, seismic resistance, environmental post-accident conditions etc.

The installation of seismic instrumentation for EU-APWR Standard Design is limited to the inside of the building in order to monitor the seismic response of the safety related components that are all installed inside the Pre-stressed Concrete Containment Vessel, Reactor Building and Power Source Building. The seismic instrumentation for the free-field motion monitoring is a voluntary undertaking of the utilities in Japan. The main reason for installing the free-field motion monitoring system is the determination of the cumulative absolute velocity parameter (CAV). This parameter will help to evaluate the potential devastation of the earthquake and decide whether restarting the plant is justified despite the operating basis earthquake (OBE) determined by the basemat monitoring having been exceeded. Free-field motion monitoring is therefore not as much

Copyright © 2015 by JSME

NII-Electronic Library Service

The Japan Society of Mechanical Engineers

required from the safety point of view, but all the more for operability.

The refuelling machine and decontamination room have no permanent radiation monitoring equipment.

During refuelling modes, mobile equipment will be used to monitor the exposure rate, including neutron dose rates, to be measured in the reactor building, at the surface of the Reactor Cavity and at the spent Fuel pit so as to detect lack of biological protection. However, this mobile equipment MHI refers to is limited to personnel dosimeters only.

The background to MHI's approach is that the requirements concerning this item widely vary between countries. Therefore, the approach is thought of as most suited for Standard Design, and it can easily be modified for specific projects.

5.3 Main NAS

A significant portion of the requirements is assigned NAS, as described in section 5 Main Results. To provide an overview of background to NAS-labels, a list is compiled showing the areas where NAS is assigned.

NAS-labels were assigned to requirements related to the following areas:

- Strategies, procedures, organisation, methods and planning related to maintenance
- Plant Life Management
- Consumables for O&M, Facilities and services to support O&M together with plant arrangements and access for O&M
- Detailed requirements on environmental impacts, both related to assessment and design features
- Construction planning, such as cost, scheduling, methods used and optimisation
- Detailed requirements on non-essential systems and components. (HVAC)
- External hazards specific to site or country
- Seismic instrumentation
- Site-specific seismic requirements
- Monitoring, inspection and testing
- Manufacturing requirements
- Material-related requirements on non-RCPB SSC
- Detailed requirements on electrical equipment
- I&C-staffing plan
- Emergency preparedness Centre
- Data management tools
- Codes & standards level IV
- Documentation
- Security
- Prime Contractors and sub-Contractors
- Decommissioning planning
- External hazards in PSA

5.4 Main BRE

All Design Extension Condition (DEC) dedicated systems are single failure tolerant. This is a design principle for diverse systems. From EUR point of view, this is beyond requirement that considers the decision to apply single failure criteria on a case-by-case basis and using best estimate methodologies.

The Emergency Power Supply (EPS), i.e. the emergency diesel generators, has enough fuel capacity stored onsite to supply power to the required loads for LOOP and LOOP+LOCA for 7 days. As 7 days is significantly more than 3 days set by EUR, the design is considered to go significantly beyond requirement.

6. CONCLUSIONS

Mitsubishi Heavy Industries' EU-APWR Standard Design was formally certified as compliant by the EUR organization in October 2014.

The two-year assessment work looked over different areas such as systems, processes, components, materials, plant layout, probabilistic safety analysis methodology, safety, perfomance and availability assessments confirming the "good compliance" of the EU-APWR Standard Design.

The requirements that are not in compliance with EUR correspond to 1.3%.

The discrepancies between the EU-APWR Standard Design and the EUR regard various aspects such as operational capability and performance, outage operation, radiation monitoring and protection, layout. None of the divergences have been considered to constitute a threat for a plant project in Europe. Some of the deviations are due to different approaches to the design philosophy and process or to differences between the rules and standards used in Japan and Europe.

The success of the assessment process was reached thanks to a tight and productive collaboration between all the parties involved: Sponsors, Supporters, Vendor and Management at all levels. The whole process contributed to increasing the knowledge of the Vendor´s design and improving different aspects of the EUR having regard to participants' feedback on the EUR-document.

REFERENCES

[1] *European Utility Requirements for LWR Nuclear Power Plants – Volume 2, Revision D*, October 2012, EUR Organization.

[2] *European Utility Requirements for LWR Nuclear Power Plants – Volume 3, Subset H – EUR Compliance Assessment of the MHI EU-APWR* Design, January 2015, EUR Organization (restricted access).

[3] Welander, D., Nuutinen, P., 2014, *EUR-APWR Project - Project Manual, Revision A, Draft 8*

Copyright © 2015 by JSME

NII-Electronic Library Service

The Japan Society of Mechanical Engineers

ABBREVIATIONS LIST

AG	Administration Group
ANSI	American National Standards Institute
BWR	Boiling Water Reactor
CAV	Cumulative Absolute Velocity
CCF	Common Cause Failure
CG	Coordination Group
DEC	Design Extension Condition
EPS	Emergency Power Supply
EUR	European Utility Requirements
EU-APWR	European version of the Advanced Pressurized Water Reactor
HVAC	Heating, Ventilation and Air Conditioning
IAEA	International Atomic Energy Agency
I&C	Instrumentation and Control
LOCA	Loss Of Coolant Accident
LOOP	Loss Of Offsite Power
MFEED	Melt Fragmentation Enhancement for Ensuring particulate Debris bed formation
MHI	Mitsubishi Heavy Industries
MOX	Mixed Oxide (fuel)
NIRMA	Nuclear Information and Records Management Association
OBE	Operating Basis Earthquake
O&M	Operation and Maintenance
PAR	Passive Autocatalytic Recombiner
PMT	Project Management Team
PSA	Probabilistic Safety Analysis
PWR	Pressurized Water Reactor
RCPB	Reactor Coolant Pressure Boundary
RWSP	Refueling Water Storage Pit
SC	Steering Committee
SGTR	Steam Generator Tube Rupture
SGWDV	Steam Generator Water Discharge Valve
SIS	Safety Injection System
SSC	Systems, Structures and Components

Copyright © 2015 by JSME

NII-Electronic Library Service